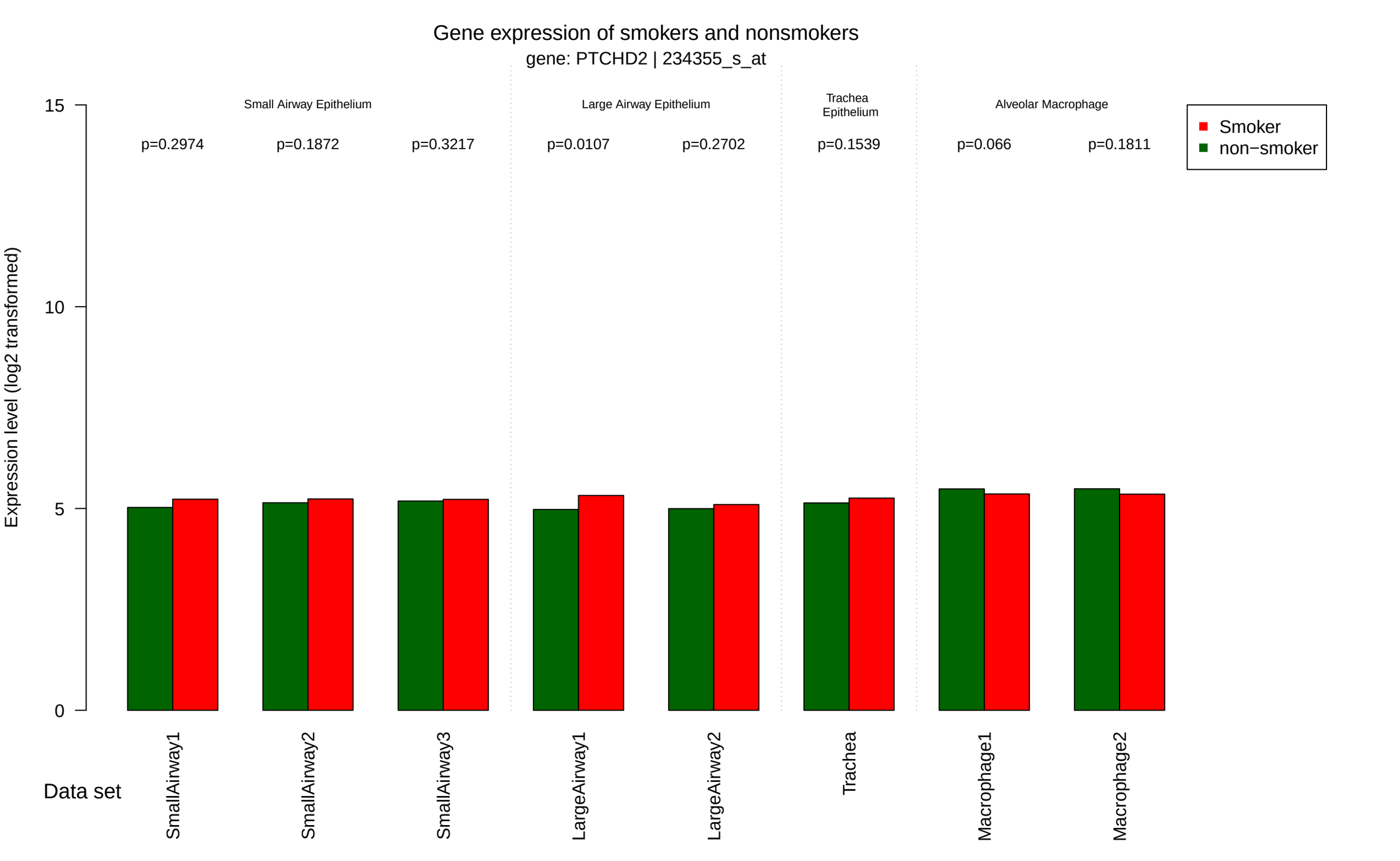
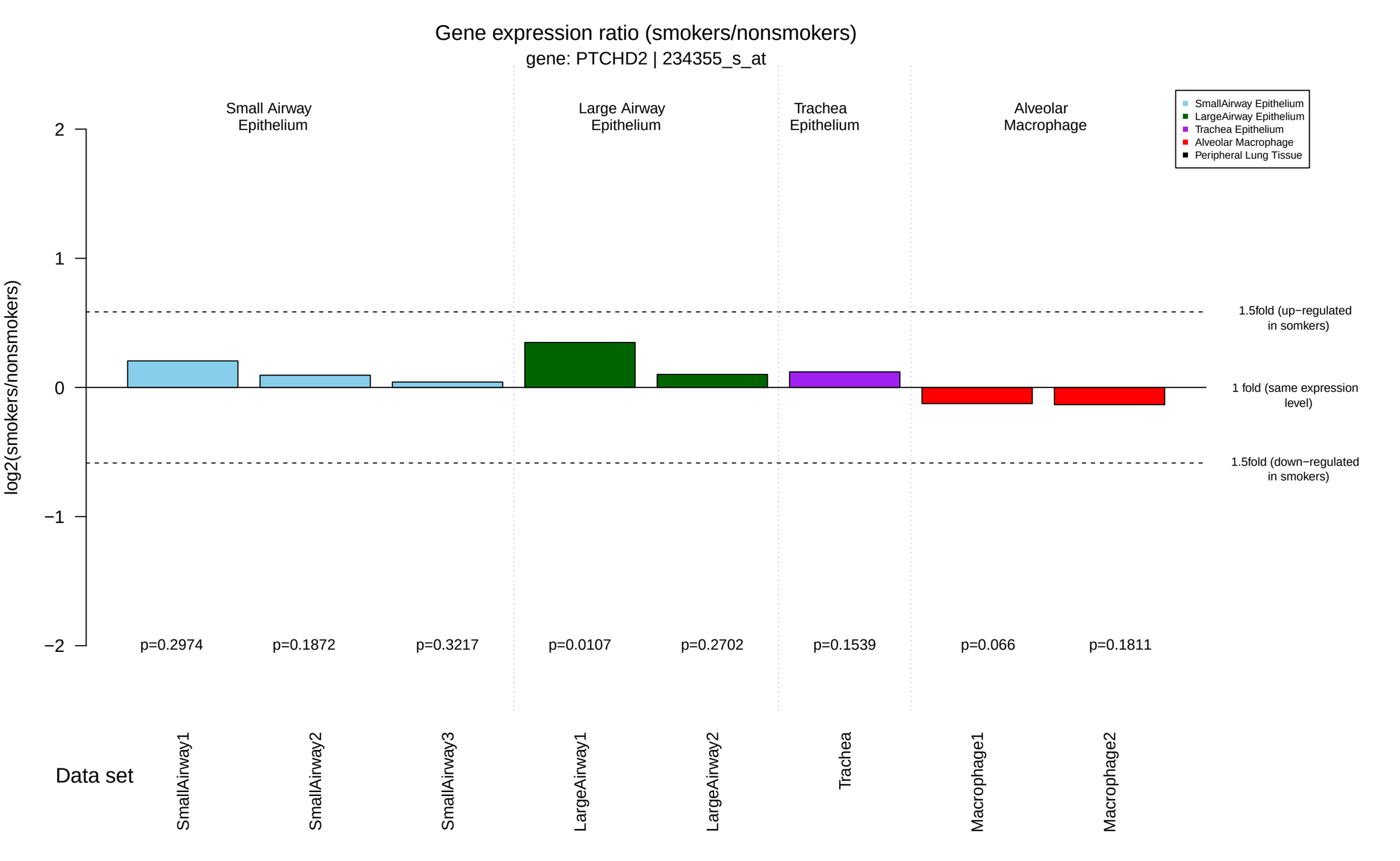
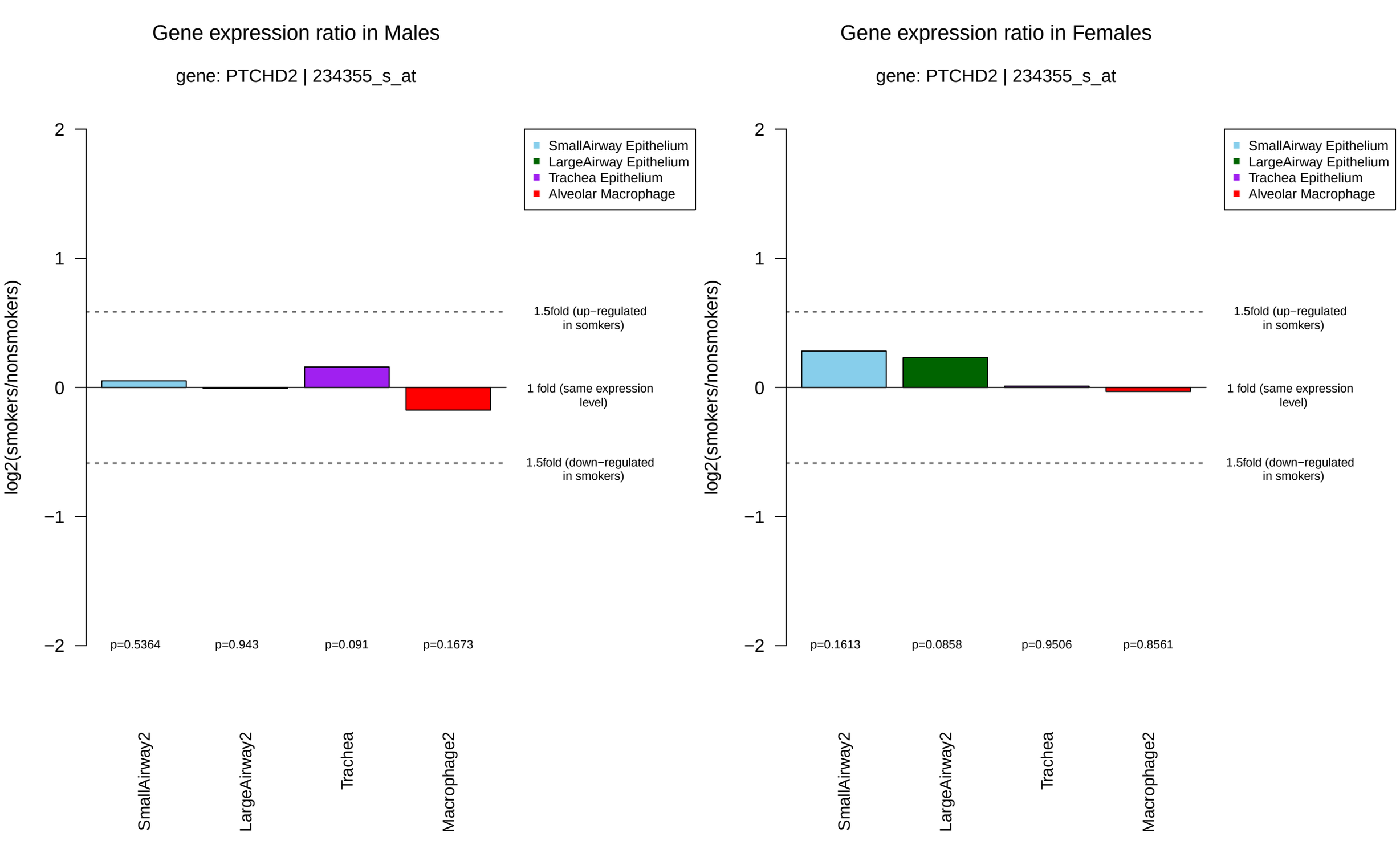
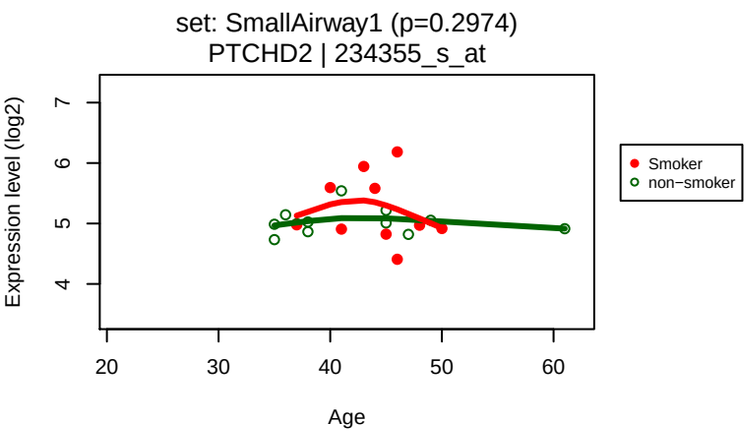
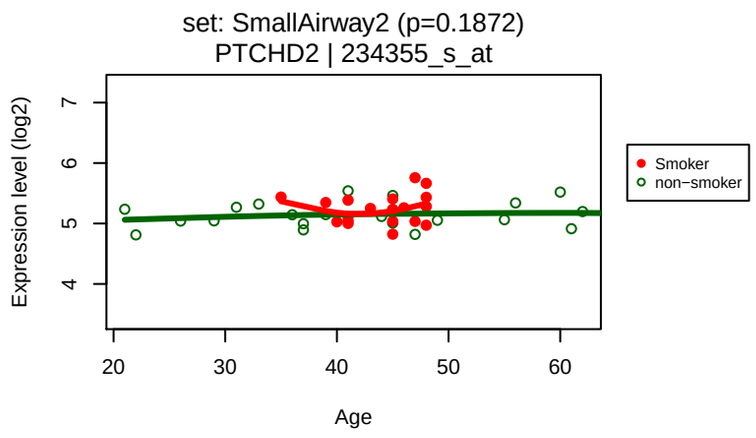
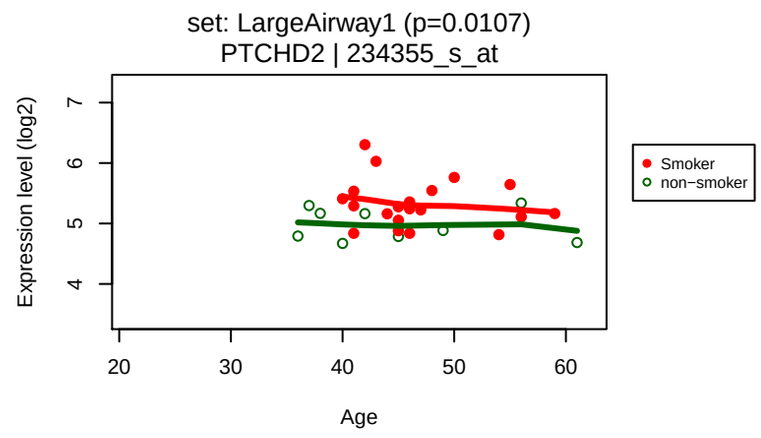

Gene expression of smokers and nonsmokers
gene: PTCHD2 | 234355_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2974
p=0.1872
p=0.3217
p=0.0107
p=0.2702
p=0.1539
p=0.066
p=0.1811
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

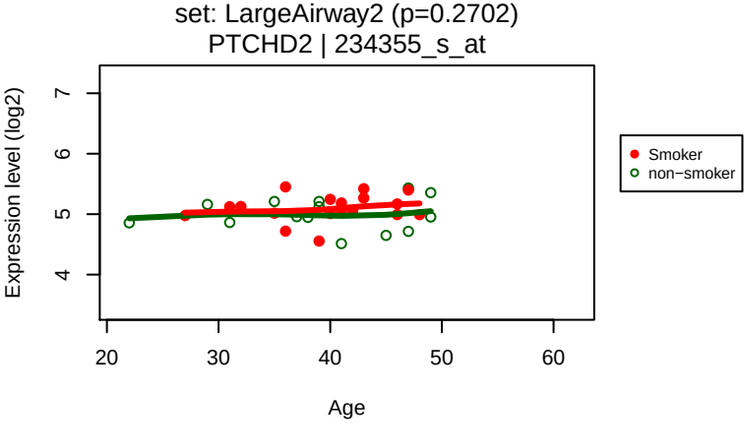

Gene expression ratio (smokers/nonsmokers)
gene: PTCHD2 | 234355_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2974
p=0.1872
p=0.3217
p=0.0107
p=0.2702
p=0.1539
p=0.066
p=0.1811
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

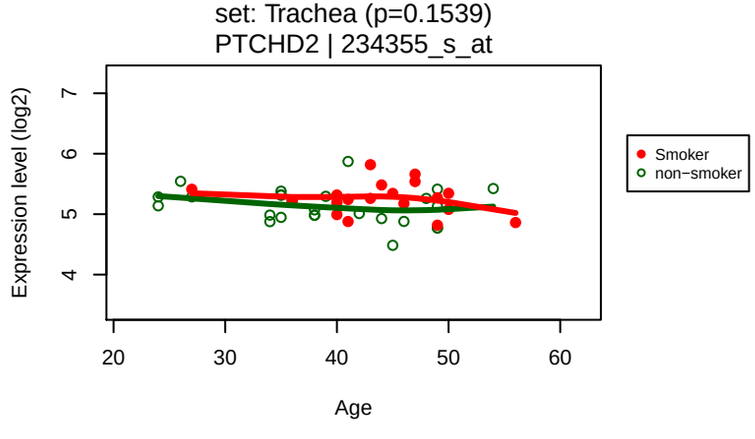

Gene expression ratio in Males
gene: PTCHD2 | 234355_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5364
p=0.943
p=0.091
p=0.1673
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PTCHD2 | 234355_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1613
p=0.0858
p=0.9506
p=0.8561
SmallAirway2
LargeAirway2
Trachea
Macrophage2

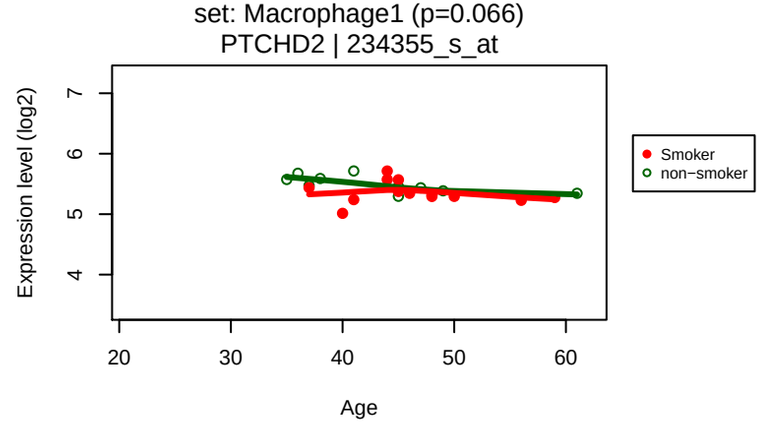

set: SmallAirway1 (p=0.2974)
PTCHD2 | 234355_s_at
Expression level (log2)
Age
Smoker
non−smoker

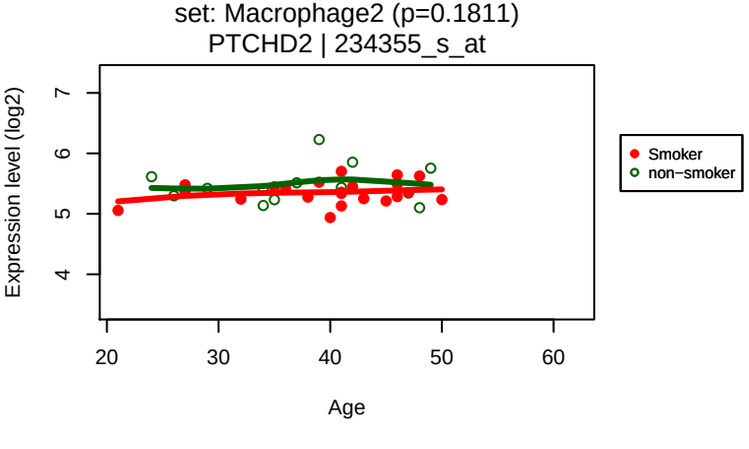

set: SmallAirway2 (p=0.1872)
PTCHD2 | 234355_s_at
Expression level (log2)
Age
Smoker
non−smoker

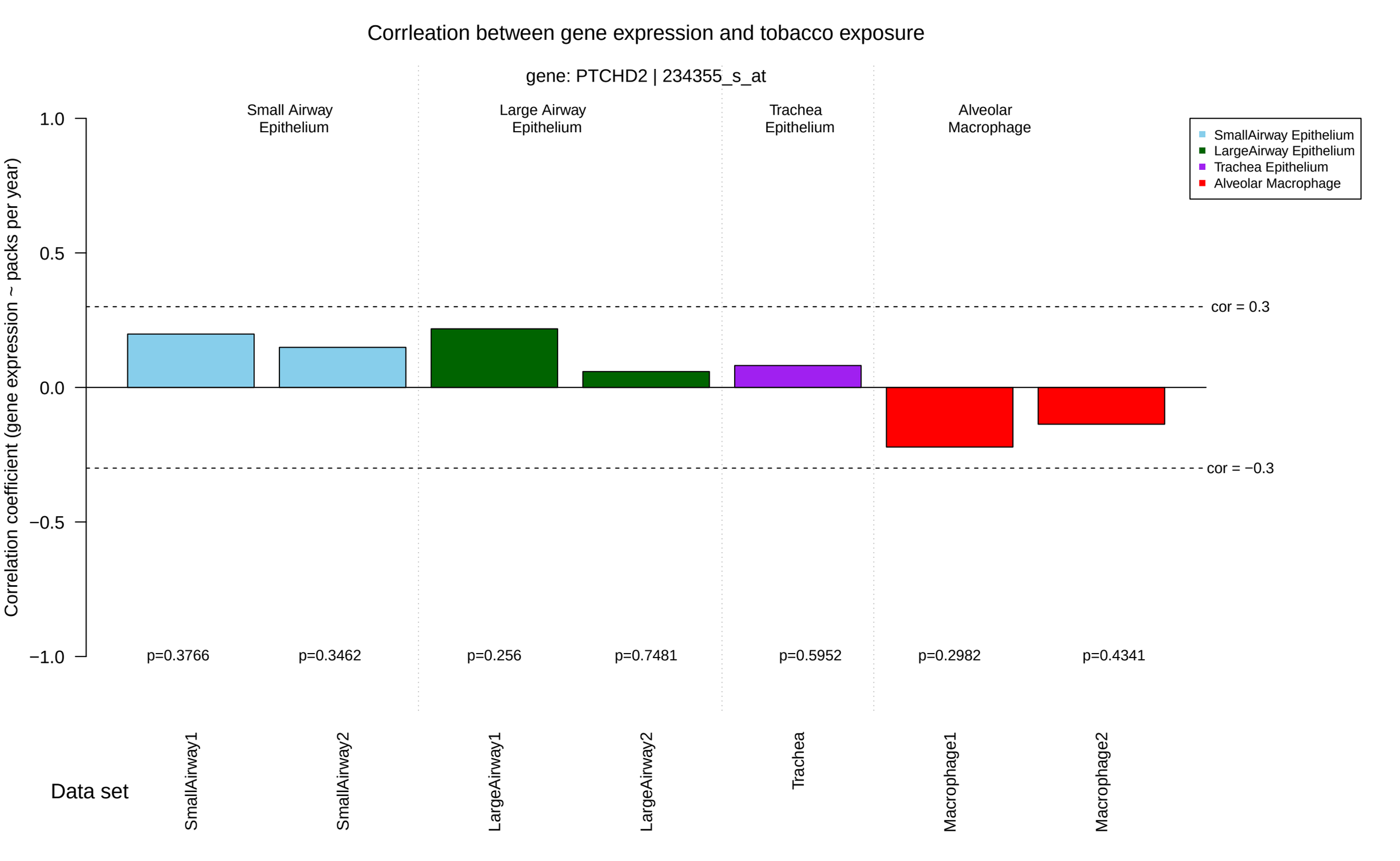

set: LargeAirway1 (p=0.0107)
PTCHD2 | 234355_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2702)
PTCHD2 | 234355_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1539)
PTCHD2 | 234355_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.066)
PTCHD2 | 234355_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1811)
PTCHD2 | 234355_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PTCHD2 | 234355_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3766
p=0.3462
p=0.256
p=0.7481
p=0.5952
p=0.2982
p=0.4341
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2